GRE Quantitative Reasoning

모의고사 Set 4 — 문제지

시범 세트 · S2 상향

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내

- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 4 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

921. x∗ is defined as the 3-digit integer formed by reversing the digits of integer x; for instance, 258∗ is equal to 852. R is a 3-digit integer such that its units digit is 2 greater than its hundreds digit.
Quantity A: R∗−R
Quantity B: 200

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

873. Certain rectangular containers can each hold a maximum of 3 cubic feet of mulch. (1 yard =3 feet)
Quantity A: The maximum number of cubic yards of mulch that 100 of these containers can hold
Quantity B: 13

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

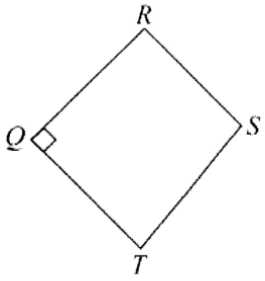
Question 3 대소비교

812. k is a positive integer, and n=15k.
Quantity A: The number of different prime factors of n
Quantity B: 4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1119.



In quadrilateral QRST, the length of RS is less than the length of QT, and RS is parallel to QT.
Quantity A: The length of QR
Quantity B: The length of ST

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 5 객관식

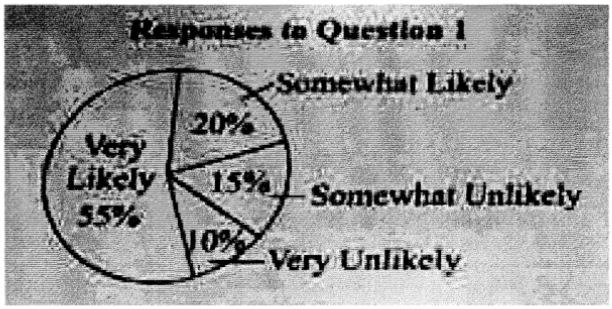
1260. How many positive integers less than or equal to 603 are multiples of 2 or multiples of 3 or both?
A.101
B.202
C.303
D.402
E.501

(A) (B) (C) (D) (E)

Questions 6 to 8 are based on the following data.

In a survey, 2,400 homeowners responded to the following question.

Question 1. How likely is it that you will continue to live in your current home for at least then more years?



The homeowners who responded "very likely" or "somewhat likely" to Question 1 were asked Question 2.

Question 2. What aspects of your current home make it likely that you will continue to live there? (Select all that apply.)

Responses to Question 2 as a Percent of Those Who Were Asked Question 2

Response	Percent
House is in a good location.	46%
House is attractive.	43%
House is comfortable.	38%
House is in good condition.	31%
House has no mortgage.	25%

Question 6 객관식

33. For how many of the five response categories in the table was the number of responses less than 750?

A. None
B. One
C. Two
D. Three
E. Four

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 7 복수선택

34. Based on the information given, which of the following statements are true?
Indicate all such statements.

A. For Question 1, the ratio of the number of homeowners who responded "somewhat likely" to the number of homeowners who responded "somewhat unlikely" is 4 to 3.

B. 75 percent of the homeowners in the survey were asked Question 2.

C. More than 40 percent of the homeowners in the survey responded "house is attractive" to Question 2.

Indicate all that apply. A B C

Question 8 객관식

35. If 24 percent of the homeowners who were asked Question 2 responded both "house is comfortable" and "home is in good condition," what percent of the homeowners who were asked Question 2 gave neither of these responses?

A. 21%

B. 31%

C. 38%

D. 45%

E. 55%

(A) (B) (C) (D) (E)

Question 9 객관식

505. Q unit of water is used to irrigate n acres of land. How many acres of land can 1000 units of water irrigate?
A Q/1000n
B n/1000Q
C 1000Q/n
D 1000n/Q
E n/Q

(A) (B) (C) (D) (E)

Question 10 주관식

434. When driving a car at a constant speed of 50 miles per hour the driver decided to apply the brake. How many feet did the car travel during the 3/4 second that elapsed between the time the driver decided to apply the brake and the time the driver actually started to apply the brake? (1 mile=5,280 feet)
_____feet

[ ] feet

Question 11 복수선택

660. In a neighborhood consisting of 2,000 homes, 80 percent of the homes are valued at $325,000 or less. Which of the following statements about the values of the homes in the neighborhood must be true?
Indicate all such statements.
AThe average (arithmetic mean) value is at most $325,000.
BThe median value is at most $325,000.
CAt most 400 homes have values greater than $325,000.

Indicate all that apply. A B C

Question 12 객관식

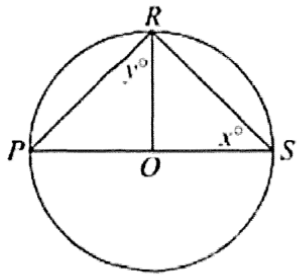
574. Machines A and B each assembled 3,000 metal clips in 15 hours and 12 hours, respectively, working alone at their constant rates. On Monday, working simultaneously at their respective constant rates, the two machines assembled a total of n clips in x hours. The next day, machine A was modified so that its constant rate decreased by 25 percent. If the two machines worked simultaneously at their respective constant rates for x hours after the modification of machine A, then the total number of clips that they assembled in x hours was approximately what percent less than it was the day before?
A6%
B11%
C16%
D20%
E25%

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

800.



Points P, R, and S are on the circle with center O. The measure of angle ROS is 92°.
Quantity A: x
Quantity B: y

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

951. In a survey of educators. 40 percent reported that they used the Internet for research. Also in the survey, 65 percent reported that they used the Internet for E-mail.
Quantity A: The percent of the educators surveyed who reported that they used the Internet for both research and E-mail
Quantity B: 26%

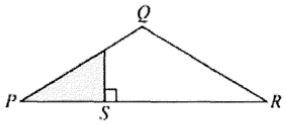
(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

935. Let m be an integer greater than 4.
Quantity A: The number of combinations of 3m objects taken m+9 at a time
Quantity B: The number of combinations of 3m objects taken 2m−9 at a time

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1237. 

PQ=QR
The area of the shaded triangular region is 1/3 the area of triangular region PQR.
Quantity A: 1/3(PR)
Quantity B: PS

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

151. When (5n+1) is divided by 3 and 4, the remainder is 2 and 1, respectively.
Quantity A: n
Quantity B: 3

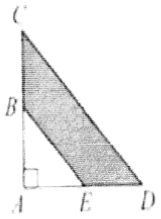
(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

385. During a 7-day period, a plant`s height increased half as much each day as it did the day before. What is the ratio of the plant`s increase in height on day 4 to its increase in height on day 7?
A1/8
B1/2
C2
D4
E8

(A) (B) (C) (D) (E)

Question 7 객관식

1074. 

In the figure, AB=AE and BC=ED=4. If the area of the shaded region is 44, what is the value of AB?
A3
B5
C7
D9
E11

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 객관식

911. From a set of 100 numbers, half were selected to form group I, and 60 percent of the remaining numbers were selected to form group II. The average (arithmetic mean) of the numbers in group I is 24.4, and the average of the numbers in group II is 31.5. Which of the following is closest to the average of the numbers in groups I and II combined?
A27.1
B27.6
C27.8
D28.0
E28.3

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 복수선택

Customer Satisfaction Ratings

Rating Information	Bank I	Bank II
Number of ratings	155	160
Mean rating	7.4	5.9
Standard deviations of ratings	1.6	1.8

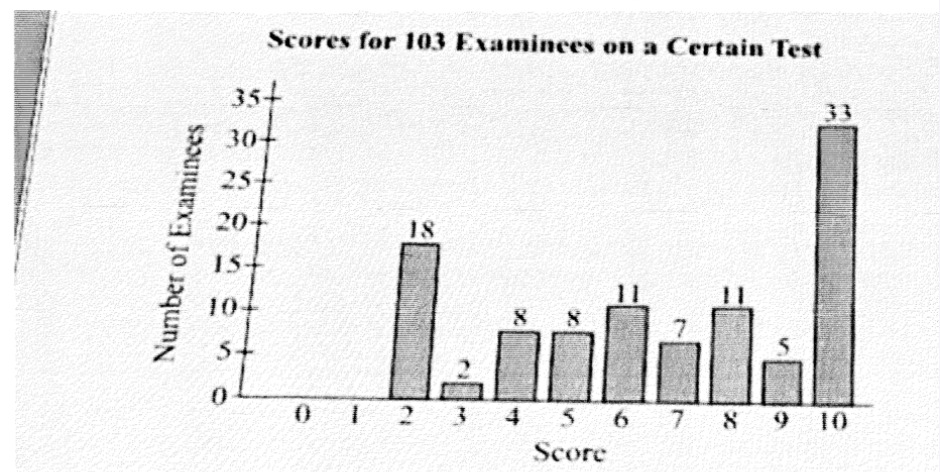
384. The table above summarizes customer satisfaction ratings for two banks, where each rating is an integer from 1 to 10. Which of the following statements must be true?
Indicate all such statements.
AFor Bank I, if a rating is within 0.5 standard deviation of the mean rating, then the rating is 7.
BFor Bank II, if a rating is within 0.4 standard deviation of the mean rating, then the rating is 6
CThe sum of all the ratings for Bank I is less than the sum of all the ratings for Bank II.

Indicate all that apply. [A] [B] [C]

Question 10 주관식

1118. x and y are both integers. $2 \leq x < y \leq 7$,what is the least possible value of (x+y)/xy?
Give your answer as a fraction.

Questions 11 to 13 are based on the following data.



The only possible scores on the test were integers between 0 and 10. No one received a score of 0 or 1. The mean score was 6.72 and the standard deviation was 2.98.

Question 11 객관식

36. Examines who received a score of 6 or more passed the test. Which of the following is closest to the average(arithmetic mean) score of the examinees who passed the test?

A. 7.5
B. 8
C. 8.5
D. 9
E. 9.5

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 12 객관식

37. What was the median score?

A. 6
B. 6.5
C. 7
D. 7.5
E. 8

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 13 객관식

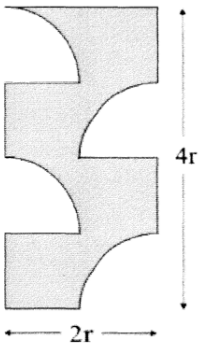
38. Which of the following is closest to the percent of the examinees who received scores that were within 1.0 standard deviation of the mean?

A. 15%
B. 40%
C. 50%
D. 90%
E. 95%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 14 객관식

784.



What is the perimeter of the shaded region of the rectangle shown above if each of the unshaded regions is a 90-degree sector of a circle with radius r?
A(12+2π)r
B(11+2π)r
C(10+2π)r
D(8+2π)r
E(4+2π)r

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 15 복수선택

610. At a certain online bookstore, the price of each book is the same. The total price of a customer order is the sum of the prices of the books in the order plus a shipping fee that is constant regardless of the number of books purchased. Last year the average (arithmetic mean) number of books per customer order was 1.5, and the average of the total prices of the orders was $10.50. Which of the following amounts could be the book price and the shipping fee last year? Indicate all such amounts.

A$4.00 per book and $4.50 shipping fee
B$4.50 per book and $3.50 shipping fee
C$5.00 per book and $3.00 shipping fee
D$5.50 per book and $2.25 shipping fee

Indicate all that apply. A B C D